

**Imagine a world where data is extremely scarce. Pick a popular product and re-think it with this constraint. How will you re-design this product, without losing its core value proposition?**

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**CONFIDENTIAL**



- **How would this world of data scarcity change user behavior and expectations?**
- **Which product or experience would you reimagine, and why that one?**
- **What core value must your product continue to deliver despite this constraint?**
- **How will you redesign the experience? What trade-offs will you make?**
- **What are the biggest risks or limitations of your approach, and how will you handle them?**
- **What does success look like? How will you measure if your redesign truly works?**



**Setting the Stage:** Our moonshot solution reimagines Google Maps by blending the **traditional human wisdom of asking locals** for directions with a device-to-device G-Mesh network **G-Mesh network** for sharing location data locally through nearby **Device-Cache** while travelling, combined with in-app optimizations to ensure seamless navigation; even in offline zones or while exploring distant regions

#1 Why Google Maps? *(chosen Product)*

Non-Negotiable Utility

Data Hungry Core Engine

Critical Infra for 100 M+ Businesses

2 Billion MAU

#3 Why Google should care?

Android Stickiness & unlock new user

Brand Trust & Goodwill

Product Resilience

Boost B2C Access & B2B Revenue

Sustainability & ESG Goals

#2 Why the Problem Matters?

Unreliable data in motion

Daily Mobility impact on Businesses

Economic

Expensive Data & Rising Costs

Economic Inequality

Social Impact

Disaster & Remote Scenarios

High Carbon Cost of Data Transfers

#4 Why NOW?

Global Data Limitation Policy

Geopolitical Push

AI/ML & IoT On-Device Maturity

Google Maps : Provides real-time navigation, traffic info, local discovery, and location-based services

Core Offerings

Navigation

Real Time Incident update

Local Business Discovery

Street / Satellite View

Offline Maps

Accuracy

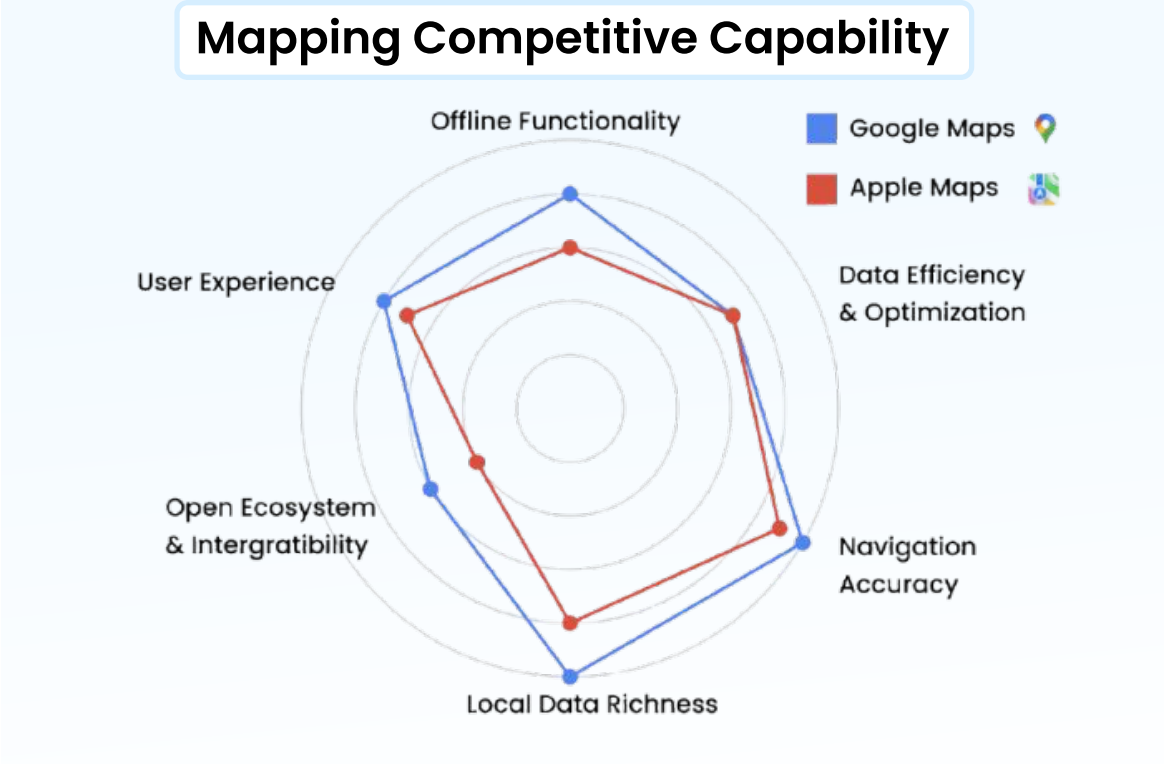
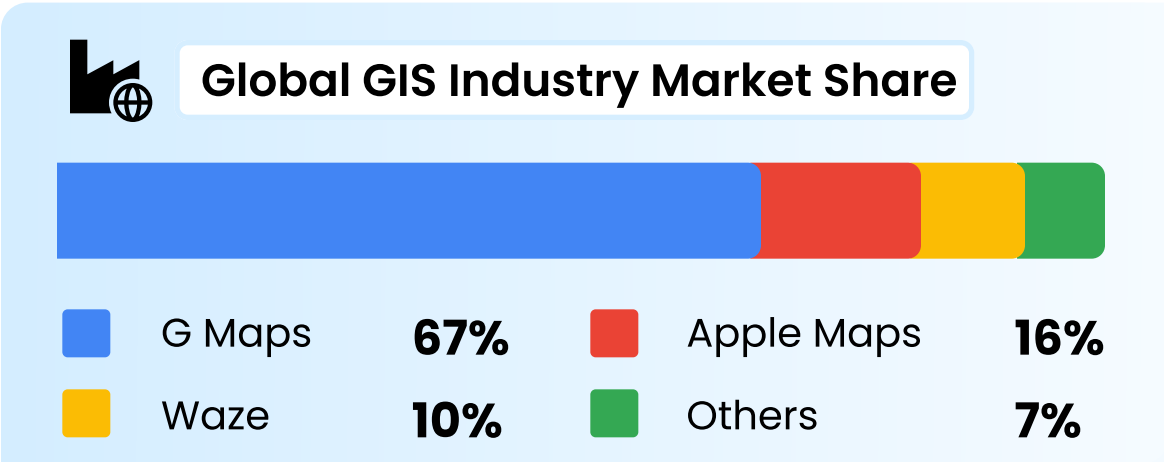
Reliability

Accessibility

User Safety

Core Values

Continued to be delivered, despite data constraints



Google Maps Vision

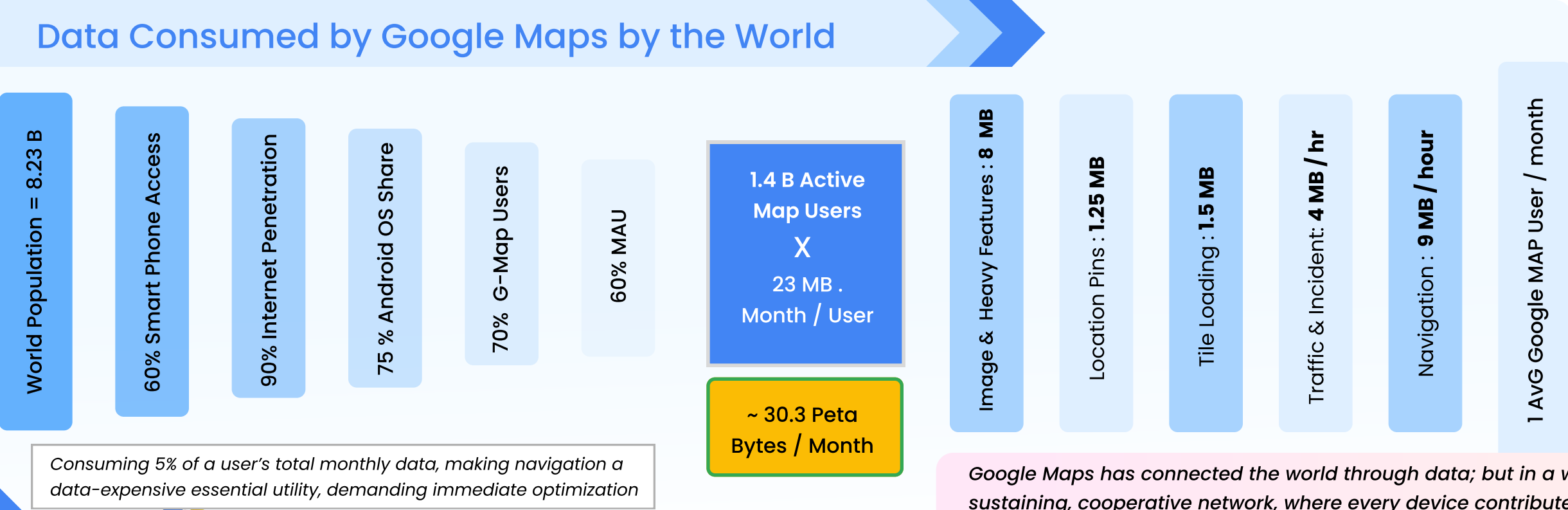
"Explore and navigate your world"



Defining the Imaginary World

Key Assumptions:

- Governments worldwide have enforced data consumption limits to reduce carbon emissions
- Android Device, current technology and human-behaviours are the same for share in this world
- Data Limit: **500 MB / Month** over the world
- Speed: 2G / 3G, approx **0.1-1 MBPS**
- Geographic Spread: **~ 50% Urban, 30% Rural**
- MAPme / G-Mesh Network Signal Carbon Footprint is **lower** than data usage for Maps







**178**  
Surveys

 **21**  
Interviews

## Surveyed Profiles

 **60%**  
Graduates

 **25%**  
Employees

### Word Cloud based on Primary Research



## Key Insights from Survey

 Users **consciously limit** their **data usage** and open Google Maps only when essential

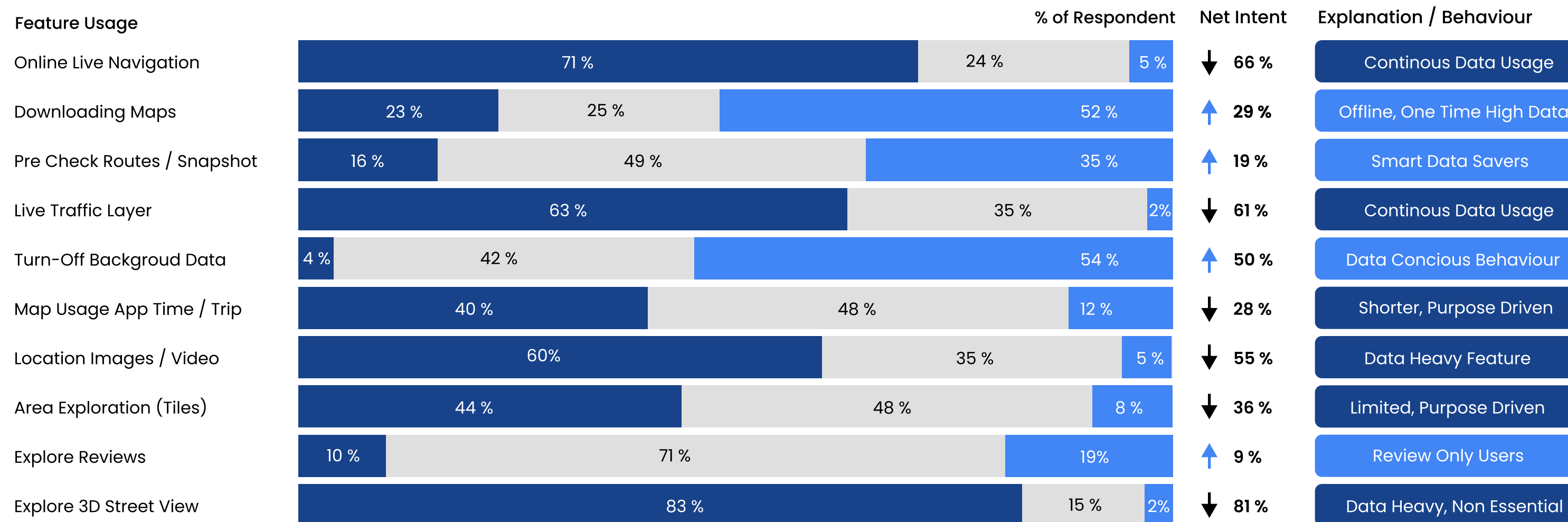
- Many users **pre-check routes**, memorize key turns, or use **screenshots** to minimize data usage

🔗 Users often **sacrifice live** update, traffic layer, and satellite view, prioritizing only **essential function**

🙄 Users would prefer to ask locals despite of opening Maps In a world of data scarcity, users need some innovation to **combine Maps with Local Knowledge**

## Navigation Behaviors Shift in a Data-Scarce World

*Data scarcity doesn't stop navigation; **it reshapes it.** Users are moving from continuous connectivity to conscious efficiency, demanding products that do more with less*



*In a data-scarce world, users become innovators adapting optimization, caching, and co-navigating. It's time for Google Maps to mirror that ingenuity.*

## User Personas

*Data-Conscious Urban Student / Prepaid packs on limited balance*

**Marie | 19 yrs | Raipur**



| Goal                                                                                | Behaviour                                                                           |                                                                                     |                                                                                     | Pain Points                                                                         |                                                                                       | Needs                                                                                 |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|  |  |  |  |  |  |  |
| Navigate new areas on minimal data                                                  | Searches locations, avoids full navigation                                          | Prefers Asking Locals where possible                                                | Disables layers to save data                                                        | Offline maps use huge storage                                                       | Reconnecting prompts waste data & time                                                | Lite map showing key details only                                                     |

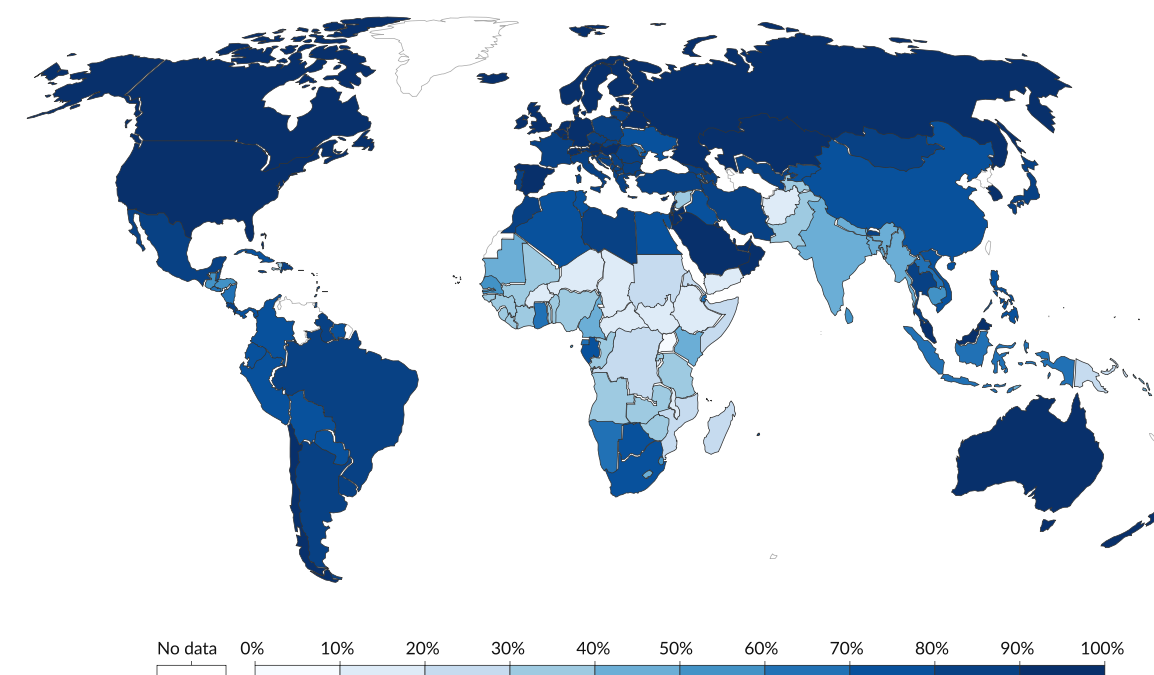
**Naveen | 25 yrs | Patna**

*Low-Tech Rural Driver operating in tier-1 zones*

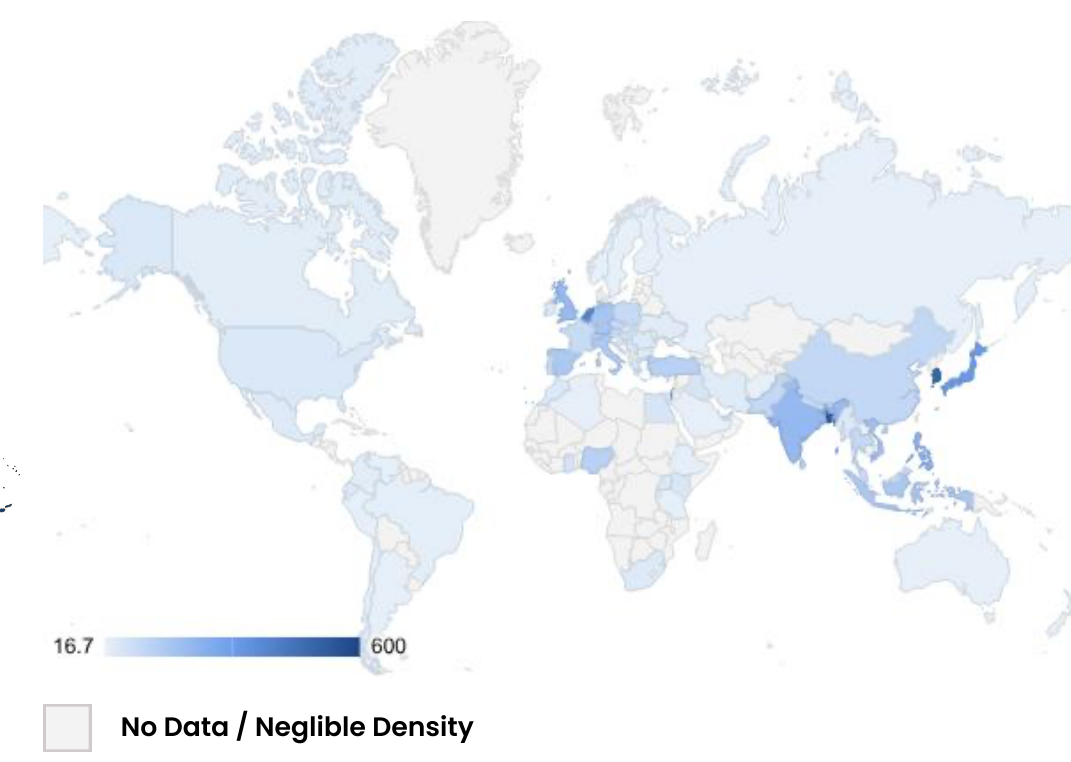


| Goal                                                                                | Behaviour                                                                           |                                                                                     |                                                                                     | Pain Points                                                                           |                                                                                       | Needs                                                                                 |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|  |  |  |  |  |  |  |
| Navigate reliably with weak signal                                                  | Uses cached or remembered routes                                                    | Switches SIMs, uses 2G fallback                                                     | Keeps battery-saver always on                                                       | Maps freeze during signal loss                                                        | High battery drains from reloads                                                      | Offline map syncing only when needed                                                  |

## World Internet Penetration



## Global Android Device Density



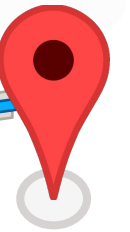
Internet Penetration and Android Device Density in any area  $> 15 / \text{sq.km}$  is crucial for creating P2P G-Mesh Network effectively





START

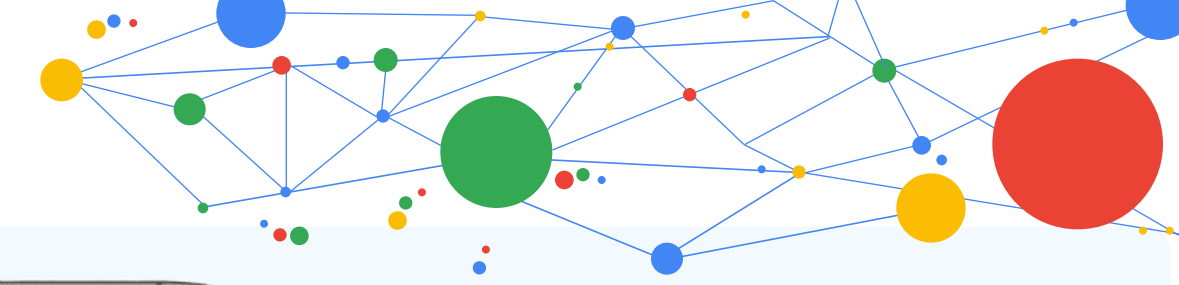
 **51 min**  
20.7 km



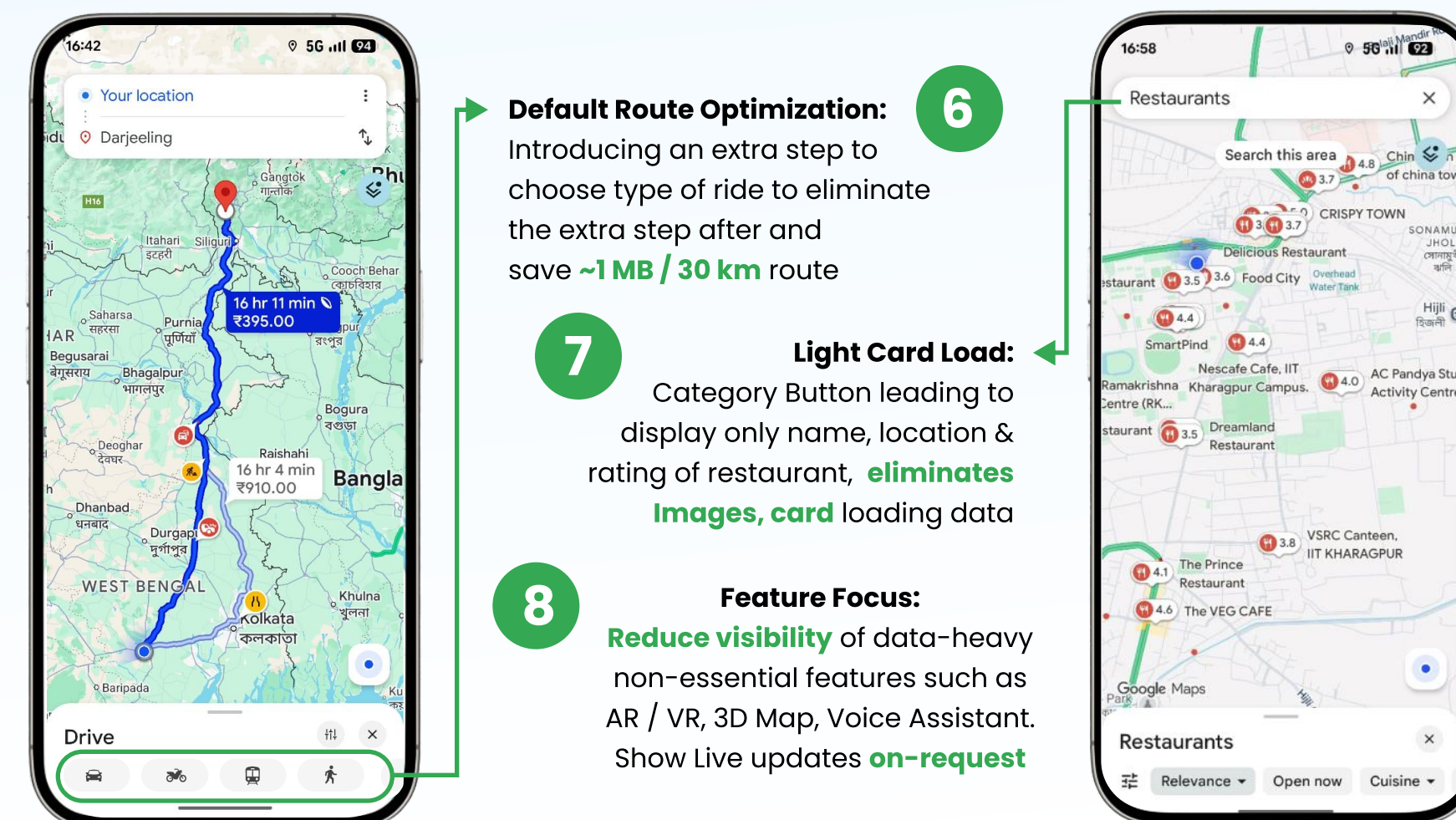




# Google Maps App Optimization & Feature Enhancement



Core User Journey : Solution Focus   ▶ Open Maps   ▶ Explore Location   ▶ Explore Category Location   ▶ Location Detail   ▶ Directions / Start   ▶ Navigate







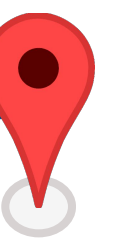
The pitchment solution saves  
**35% (8 MB of 23 MB) / Month / User**  
***Equivalent to 10 Peta Byte globally***

**Assumption :**

- A user engages 3 x 1 hour session navigating / exploring Maps,
- 2 x navigation session travelling around 40 km;
- 1 x distant exploration session;
- explores 20 sq.km, pins, images, etc

Frequent Riders & Passive Users are balanced out at average numbers.

| RICE Feature / Solution Prioritization |       |        |            |        |       |      |
|----------------------------------------|-------|--------|------------|--------|-------|------|
| Feature                                | Reach | Impact | Confidence | Effort | Score | Rank |
| Minimalist Map                         | 10    | 6      | 10         | 4      | 150   | #3   |
| Essential Pin Filter                   | 9     | 4      | 9          | 5      | 64.8  | #8   |
| Range Meter                            | 8     | 4      | 7          | 5      | 44.8  | #10  |
| Lazy Image Load                        | 6     | 9      | 10         | 4      | 135   | #6   |
| Personalized AI                        | 6     | 8      | 8          | 6      | 64    | #9   |
| Route Flow Optimization                | 9     | 7      | 10         | 2      | 315   | #1   |
| Category Card Optimization             | 6     | 8      | 9          | 3      | 144   | #5   |
| Feature Focus                          | 9     | 8      | 8          | 4      | 144   | #4   |
| G-Mesh Research                        | 10    | 9      | 10         | 5      | 180   | #2   |
| G-Mesh Development                     | 8     | 9      | 10         | 8      | 90    | #7   |







| Metric Type        | Metric                         | Description                                              | Goal                           |
|--------------------|--------------------------------|----------------------------------------------------------|--------------------------------|
| ★ North Star       | Low-data navigations / user    | Completed low-data trips (<2 MB) per active user         | 80% trips completed <2 MB      |
| Level 1 Metrics    | Data per Session               | Average MB consumed per completed trip                   | Reduce avg session to < 1.5 MB |
|                    | Trip Success Rate              | % of users reaching destination without disruption       | 90% users                      |
|                    | Lite/Mesh MAU                  | Total unique users leveraging new low-data features      | 1.5 Billion MAUs globally      |
| Level 2 Metrics    | Caching Performance            | % of map tiles served from the device cache              | 60% of tile data               |
|                    | Peer-to-Peer Mesh Efficiency   | % of map/route bytes served by P2P/mesh network          | 70% bytes                      |
|                    | Offline Mode Utilization       | % navigation sessions fully completed entirely offline   | 50% Navigations                |
|                    | Reroute Success Rate (Offline) | % of local reroutes successful without a live connection | 40% reroutes resolved < 5 s    |
| Supporting Metrics | Latency Delta vs Online Mode   | Offline/mesh load time difference compared to online     | < +1.5 seconds lower           |
|                    | Battery Impact per Session     | Extra battery used in mesh/offline mode                  | ~20–25 mAh per 30-min trip     |
|                    | Opt-in Rate for Mesh Network   | % of eligible users actively enabling P2P mesh opt-in    | 50% devices                    |
|                    | Map Update Propagation Delay   | Time taken for local map edits to sync nearby            | < 5 minutes                    |

| Possible Pitfalls and How to overcome them:                                         |                                                                                     |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |                                                                                       |                                                                                       |                                                                                       |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Technical & Infrastructure Risks                                                    |                                                                                     | Data Accuracy & Freshness Risks                                                     |                                                                                      | User Adoption & Behavior Risks                                                        |                                                                                       | Privacy & Security Risks                                                              |                                                                                       | Business & Scalability Risks                                                          |                                                                                       |
| Building reliable, low-data mesh systems  hard with bandwidth limit                 |                                                                                     | Limited data exchange causes stale routes and outdated POIs                         |                                                                                      | Users may resist sync/sharing due to privacy or battery fears                         |                                                                                       | P2P sharing and local storage risk data leaks/violations                              |                                                                                       | costs, unclear offline monetization, ecosystem fragmentation                          |                                                                                       |
|  |  |  |  |  |  |  |  |  |  |
| Start with mesh rollout in dense regions first                                      | Use compression, edge AI, and lightweight vector tiles                              | Sync critical roads and business listings first                                     | Use on-device ML to predict route changes                                            | Onboard by creating privacy & battery safety awareness                                | Reward users for helping others navigate offline                                      | Encrypt all mesh broadcast data by default                                            | Share only anonymized, non-identifiable map tiles                                     | Partner with telcos/OEMs for OS-level mesh integration                                | Launch MVP in developing regions before scaling                                       |

● Current Data Rich Map    ● Post Solution Implement  
○ Closeness to metric define tradeoff weightage

